

decreasing latency and increasing power efficiency. The streamlined topology is particularly beneficial for workloads where every microsecond matters, such as training transformer-based language models.

The XCDs are built on TSMC's N3P process node, and the IODs use the more mature 6nm process. This pairing reflects a strategy that balances leading-edge performance with manufacturability and cost. Architecturally, CDNA 4 also includes revamped compute units optimised for low-precision types like FP4 and FP6, positioning MI350 well for training and inference on large models.

Chiplet and packaging innovations

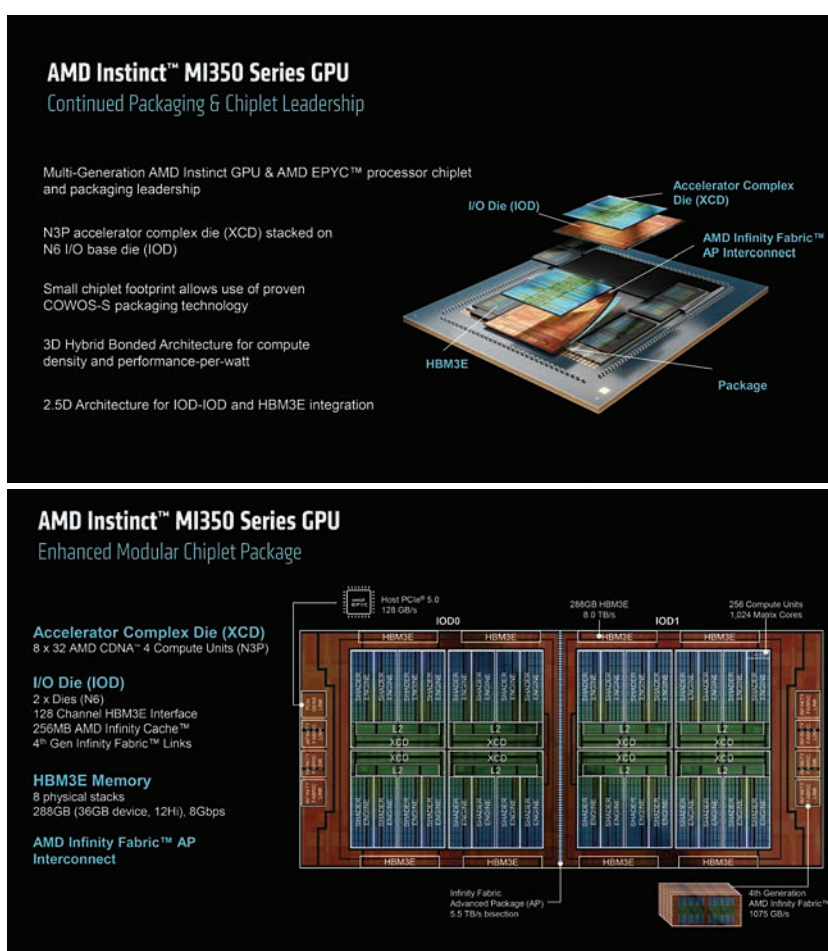
AMD's prowess in chiplet design comes in focus with the MI350X. The company leans into advanced packaging techniques like CoWoS-S (Chip-on-Wafer-on-Substrate - Stacked) and SoIC (System on Integrated Chip), enabling vertical stacking of the XCDs directly atop the IODs. This high-density stacking helps deliver more compute in a smaller footprint while improving interconnect bandwidth between compute and I/O layers.

The MI350X integrates eight XCDs, each with 32 compute units, alongside eight HBM3E memory stacks and a 256 MB Infinity Cache. The choice to use TSMC's N3P for the compute dies allows for increased transistor density, while the 6nm IODs handle memory control, I/O functions, and chip-to-chip communication. The combined architecture supports higher yields and thermal efficiency compared to a monolithic die of the same scale.

More than performance, this approach reflects a shift in data centre GPU design where modularity, stacking, and heterogeneous integration are critical for scalability and manufacturing viability.

Compute Engine and core configuration

Each XCD in the MI350X has 32 com-



pute units, a total of 256 CUs across the chip. This makes MI350X one of AMD's most compute-dense accelerators ever. It's not just raw numbers. The compute units have been redesigned for better AI throughput, particularly for matrix-heavy operations essential in machine learning.

Support for ultra-low precision formats such as FP6 and FP4 is a defining feature of this generation. These datatypes are especially valuable in generative AI models, where memory pressure and throughput constraints are challenges that constantly persist. The inclusion of these formats enables MI350 to perform more operations per second while keeping energy and bandwidth demands in check.

Changes to the SIMD lanes, vector pipelines, and matrix cores ensure that each CU is better aligned with the needs of AI, while still retaining

capabilities for traditional HPC tasks like FP64 simulations. This dual-use flexibility allows MI350 to be deployed in both AI and scientific computing contexts without compromise.

Memory subsystem: HBM3E and Infinity Cache

AMD has equipped the MI350X with 288 GB of HBM3E memory across eight stacks. This configuration delivers a total bandwidth of up to 8 terabytes per second, one of the highest in the industry today. Such high memory capacity and throughput allow the MI350X to run massive language models, including up to 520 billion parameters, without the need to split them across multiple GPUs. **[d]**

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